CS5275 The Algorithm Designer's Toolkit
S2 AY2025/26

Final Exam **Solution**

5:00 pm – 7:00 pm (2 hours), April 30 (Thursday)

Instructions:

1. **Do not** turn over any page of this paper until instructed to begin.
2. This is an **open-book** assessment; however, the use of calculators or any electronic devices is **not permitted**.
3. The exam consists of 13 true/false questions, 3 multiple-choice questions, and 3 proof-based questions, for a total of 25 marks (worth 40% of the grade).
4. Please write your answers on the answer sheet.
5. All graphs are assumed to be simple (i.e., without self-loops or multiple edges) and undirected.
6. A Boolean function $f : \{-1,1\}^n \to \{-1,1\}$ is **constant** if ($f(x) = 1$ for all $x \in \{-1,1\}^n$) or ($f(x) = -1$ for all $x \in \{-1,1\}^n$).
7. In the definition of mixing time $T_{\mathrm{mix}}(G, \epsilon)$, we consider **lazy random walks**.
8. In True-or-false Questions 12–13, we consider the **specific** multiplicative weights update algorithm for a finite two-player zero-sum game discussed in class.

True or False (13 marks: 1 mark per question)

Which of the following statements are correct?
For each statement, indicate whether it is **true** (T or ◯) or **false** (F or ×).

1. If a graph G has diameter $\Omega(\sqrt{n})$, then the conductance of G is $O\left(\frac{\log n}{\sqrt{n}}\right)$. ◯
2. $T_{\text{mix}}(G, 0.01) \in O(1)$ implies $\Phi(G) \in \Omega(1)$. ◯
3. Select two vertices u and v independently and uniformly at random (with replacement) from any 6-regular n-vertex Ramanujan graph. Then $\mathbb{E}[\mathsf{dist}(u, v)] \in o(\log n)$. ×
4. There exists a Boolean function $f : \{-1, 1\}^n \to \{-1, 1\}$ whose distance to every parity function $\{\chi_S : S \subseteq [n]\}$ is exactly $1/2$. ×
5. There exists a Boolean function $f : \{-1, 1\}^n \to \{-1, 1\}$ such that $\hat{f}(S) \neq 0$ for all $S \subseteq [n]$. ◯
6. Every monotone and odd Boolean function $f : \{-1, 1\}^n \to \{-1, 1\}$ is unanimous. ◯
7. In a weighted majority function
$$f(x) = \operatorname{sgn}(a_0 + a_1 x_1 + \cdots + a_n x_n),$$
if $a_i \neq 0$ for every $i \in [n]$, then every voter i has positive influence, i.e., $\mathrm{Inf}_i[f] > 0$. ×
8. For any Boolean function $f : \{-1, 1\}^n \to \{-1, 1\}$ and any $\epsilon \in (0, 1)$, there exists a collection $\mathcal{F}$ of subsets of $[n]$ with $|\mathcal{F}| \in n^{O(1)} \cdot \epsilon^{-O(1)}$ such that the Fourier spectrum of f is ϵ-concentrated on $\mathcal{F}$. ×
9. For any Boolean function $f : \{-1, 1\}^n \to \{-1, 1\}$, if $\mathrm{Inf}_i[f] \leqslant \epsilon/n$ for every $i \in [n]$, then f is $O(\epsilon)$-close to a constant function. ◯
10. For any Boolean function $f : \{-1, 1\}^n \to \{-1, 1\}$, if f is ϵ-close to a constant function, then $\mathrm{Inf}_i[f] \in O(\epsilon/n)$ for every $i \in [n]$. ×
11. In class, we showed that for any deterministic algorithm for learning from expert advice with $n = 2$ experts, an adversary can force the number of mistakes to satisfy $M \geqslant 2 \min_{i \in [n]} M_i$. The same lower bound $M \geqslant 2 \min_{i \in [n]} M_i$ also holds when there are $n = 10$ experts. ◯
12. In the analysis of the multiplicative weights update algorithm for a finite two-player zero-sum game discussed in class, we constructed a near-optimal row mixed strategy and showed that there exists $t \in [T]$ such that
$$\max_j A\big(\mathbf{p}^{(t)}, j\big) \leqslant \lambda^* + \epsilon.$$
Suppose instead that t is chosen uniformly at random from $[T]$. Then
$$\mathbb{E}\left[\max_j A\big(\mathbf{p}^{(t)}, j\big)\right] \leqslant \lambda^* + \epsilon.$$ ◯
13. In the same analysis, we constructed a near-optimal column mixed strategy. Specifically, for the distribution $\tilde{\mathbf{q}}$ defined by
$$\tilde{\mathbf{q}}(j) = \frac{|\{t : j^{(t)} = j\}|}{T},$$
we showed that
$$\min_i A(i, \tilde{\mathbf{q}}) \geqslant \lambda^* - \epsilon.$$
Suppose instead that t is chosen uniformly at random from $[T]$. Then
$$\mathbb{E}\left[\min_i A\big(i, j^{(t)}\big)\right] \geqslant \lambda^* - \epsilon.$$ ×

Multiple-Choice Questions (3 marks: 1 mark per question)

For each of the following questions, **select the correct answer** from the five choices.
Each question has exactly one correct answer.

1. What is the conductance of an n-vertex path, where n is even?

   (a) $2/n$
   (b) $2/(n-1)$
   (c) $1/n$
   (d) $1/(n-1)$ Answer
   (e) None of the above

2. What is the mixing time $T_{\text{mix}}(G, 0.01)$ of an n-vertex star graph ($n \geqslant 3$)?

   (a) $\Theta(1)$ Answer
   (b) $\Theta(\log \log n)$
   (c) $\Theta(\log n)$
   (d) $\Theta(n)$
   (e) None of the above

3. In the problem of learning from expert advice, we showed in class that when there is a perfect expert, there exists a deterministic strategy that makes at most $\log n$ mistakes. Now consider a setting in which a $\Theta\big(1/\sqrt{\log n}\big)$ fraction of the n experts are perfect. What is the worst-case number of mistakes made by the best possible algorithm in this setting?

   (a) $\Theta(1)$
   (b) $\Theta(\log \log n)$ Answer
   (c) $\Theta(\sqrt{\log n})$
   (d) $\Theta(\log n)$
   (e) None of the above

Proof-Based Questions (9 marks: 3 marks per question)

1. Let $G = (V, E)$ be a graph with minimum degree δ. Suppose $(S, V\backslash S)$ is a cut with conductance

$$\Phi(S) \leqslant \tfrac{1}{2}.$$

Prove that both sides of the cut must be large, namely,

$$|S| \in \Omega(\delta) \qquad \text{and} \qquad |V\backslash S| \in \Omega(\delta).$$

Proof. Since $\Phi(S) \leqslant \frac{1}{2}$,

$$|E(S, V\backslash S)| \leqslant \frac{1}{2} \min\{\mathrm{vol}(S), \mathrm{vol}(V\backslash S)\}.$$

If $|S| < \delta/2$, then every $v \in S$ has at most $|S| - 1 < \delta/2 \leqslant \deg(v)$ neighbors inside S, so

$$|E(S, V\backslash S)| > \frac{1}{2} \mathrm{vol}(S),$$

contradicting the conductance bound. Hence $|S| \geqslant \delta/2$.
The same argument also shows that $|V\backslash S| \geqslant \delta/2$. □

2. Let $f : \{-1, 1\}^n \to \{-1, 1\}$ be an unknown Boolean function that is both **monotone** and **symmetric**. You may query f at any point $x \in \{-1, 1\}^n$.

Design a *deterministic* algorithm that determines f in the fewest number of queries in the worst case. Prove that your algorithm is *correct* and *optimal*.

Proof. Since f is symmetric, its value depends only on the number of 1's in the input. Let $k \in \{0, 1, 2, \dots, n\}$ be this number. Since f is monotone, there is a threshold

$$t \in \{0, 1, \dots, n + 1\}$$

such that

$$f(x) = \begin{cases} -1, & k < t, \\ 1, & k \geqslant t. \end{cases}$$

So f is completely determined by t, and there are exactly $n + 2$ possible functions.

Algorithm: Do a binary search for t = the smallest k with value 1. A query of f on x with k 1's tells us whether $t \leqslant k$ or $t > k$, so each query halves the set of possible thresholds. Thus the algorithm correctly determines t, and hence f, in

$$\lceil \log_2(n + 2) \rceil$$

queries in the worst case.

Optimality: There are $n+2$ possible functions, but a deterministic algorithm using q queries has at most 2^q possible outputs. Hence

$$2^q \geqslant n+2,$$

so every deterministic algorithm needs

$$q \geqslant \lceil \log_2(n+2) \rceil$$

queries in the worst case. □

3. Consider a variant of the **deterministic weighted majority algorithm** for learning from expert advice with n experts using **exponential updates**.

   Initially, all experts have a weight of $w_i^{(1)} = 1$. At each round t, the algorithm predicts up/down according to the weighted majority vote of the experts. After observing the correct outcome, the weights are updated as follows: for every expert i that predicts incorrectly, its weight is updated by

   $$w_i^{(t+1)} = w_i^{(t)} \cdot e^{-\eta},$$

   where $\eta > 0$ is a fixed parameter, while the weights of correct experts remain unchanged.

   Let M denote the total number of mistakes made by the algorithm, and let M_i denote the number of mistakes made by expert i. Prove that

   $$M \leqslant \frac{M_i\eta + \ln n}{\ln\left(\frac{2}{1+e^{-\eta}}\right)}.$$

Proof. Let

$$\Phi^{(t)} = \sum_{j=1}^{n} w_j^{(t)}.$$

Initially $\Phi^{(1)} = n$.

Whenever the algorithm makes a mistake, at least half of the total weight was on experts that were wrong. Thus after the update,

$$\Phi^{(t+1)} \leqslant \frac{1}{2}\Phi^{(t)} + \frac{1}{2}e^{-\eta}\Phi^{(t)} = \frac{1+e^{-\eta}}{2}\,\Phi^{(t)}.$$

If the algorithm makes M mistakes total, then

$$\Phi^{(T+1)} \leqslant n\left(\frac{1+e^{-\eta}}{2}\right)^M.$$

Now fix expert i. Each time expert i is wrong, its weight is multiplied by $e^{-\eta}$, so after T rounds,

$$w_i^{(T+1)} = e^{-\eta M_i}.$$

Since $w_i^{(T+1)} \leqslant \Phi^{(T+1)}$,

$$e^{-\eta M_i} \leqslant n\left(\frac{1+e^{-\eta}}{2}\right)^M.$$

Taking logarithms,

$$-\eta M_i \leqslant \ln n + M \ln\left(\frac{1+e^{-\eta}}{2}\right).$$

Rearranging,

$$M \leqslant \frac{M_i\eta + \ln n}{\ln\left(\frac{2}{1+e^{-\eta}}\right)}.$$ □

— End of the exam paper —